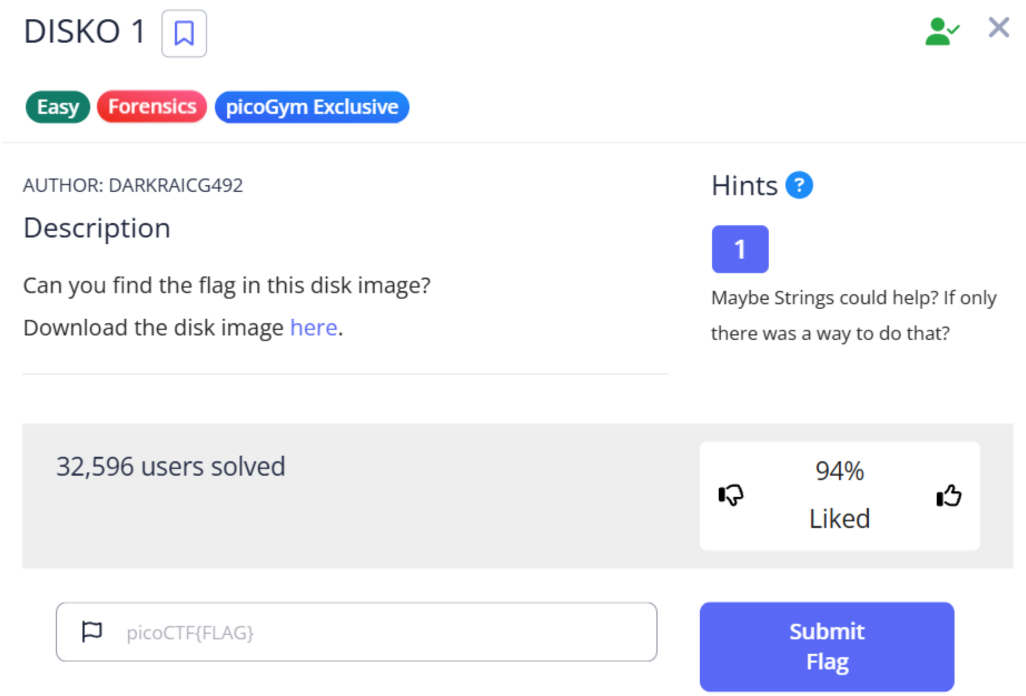
HERNANDO, PANTINE B. **BSIT-3**

PICO CTF-CHALLENGE 3



Disk Forensics Challenge

Challenge: Find the hidden flag in a disk image.
File: disko-1.dd.gz (~20 MB compressed, ~50 MB uncompressed)

Steps Taken

1. ***Download the file***

wget

https://artifacts.picoctf.net/c/536/disko-1.dd.gz

2. ***Verify the file type***

file disko-1.dd.gz

Output: gzip compressed data, confirms it is a compressed disk image.

3. ***Stream decompression and extract readable text***

zcat disko-1.dd.gz | strings | grep pico

4. ***Retrieve the flag***

picoCTF{1t5_ju5t_4_5tr1n9_c63b02ef}

```
Pantine23-picoctf@webshell:~$ wget https://artifacts.picoctf.net/c/536/disko-1.dd.gz
--2026-03-11 16:28:59--  https://artifacts.picoctf.net/c/536/disko-1.dd.gz
Resolving artifacts.picoctf.net (artifacts.picoctf.net)... 3.170.131.72, 3.170.131.33, 3.170.131.18, ...
Connecting to artifacts.picoctf.net (artifacts.picoctf.net)|3.170.131.72|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 20484476 (20M) [application/octet-stream]
Saving to: 'disko-1.dd.gz'

disko-1.dd.gz                                   100%[=====================================================================================================================>]  19.54M  1.82MB/s    in 11s

2026-03-11 16:29:10 (1.81 MB/s) - 'disko-1.dd.gz' saved [20484476/20484476]

Pantine23-picoctf@webshell:~$ curl https://artifacts.picoctf.net/c/536/disko-1.dd.gz >> disko-1.dd.gz
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 19.5M  100 19.5M    0     0  1795k      0  0:00:11  0:00:11 --:--:-- 1853k
Pantine23-picoctf@webshell:~$ ls
 Addadshashanammu        challenge.zip     challenge.zip.3   challenge.zip.6   code.py        convertme.py.1   drop-in     file.1   file.jpg    files.zip.1   flag.txt      ltdis.sh   strings     trace.pcap.1  'trings trace.pcap.2'
 Addadshashanammu.zip    challenge.zip.1   challenge.zip.4   challenge.zip.7   codebook.txt   disko-1.dd.gz    ende.py     file.2   files       flag          flag.txt.en   pw.txt     strings.1   trace.pcap.2   warm
 README.txt              challenge.zip.2   challenge.zip.5   challenge.zip.8   convertme.py   downloads        ende.py.1   file.3   files.zip   flag.1        img.jpg       static     trace.pcap  trace.pcap.3
Pantine23-picoctf@webshell:~$ file disko-1.dd.gz
disko-1.dd.gz: gzip compressed data, was "disko-1.dd", last modified: Thu May 15 18:48:13 2025, from Unix, original size modulo 2^32 52428800
Pantine23-picoctf@webshell:~$ gzip -d disko-1.dd.gz

gzip: disko-1.dd: No space left on device
Pantine23-picoctf@webshell:~$ file disko-1.dd
disko-1.dd: cannot open `disko-1.dd' (No such file or directory)
Pantine23-picoctf@webshell:~$ file disko-1.dd.gz
disko-1.dd.gz: gzip compressed data, was "disko-1.dd", last modified: Thu May 15 18:48:13 2025, from Unix, original size modulo 2^32 52428800
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disko-1.dd.gz: gzip compressed data, was "disko-1.dd", last modified: Thu May 15 18:48:13 2025, from Unix, original size modulo 2^32 52428800
Pantine23-picoctf@webshell:~$ strings disko-1.dd | grep pico
strings: 'disko-1.dd': No such file
Pantine23-picoctf@webshell:~$ ^C
Pantine23-picoctf@webshell:~$ zcat disko-1.dd.gz | strings | grep pico
:/icons/appicon
# $Id: piconv,v 2.8 2016/08/04 03:15:58 dankogai Exp $
piconv -- iconv(1), reinvented in perl
  piconv [-f from_encoding] [-t to_encoding]
  piconv -l
  piconv -r encoding_alias
  piconv -h
B<piconv> is perl version of B<iconv>, a character encoding converter
a technology demonstrator for Perl 5.8.0, but you can use piconv in the
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```

Reflection

This challenge highlights a key forensic technique: when disk images are too large to decompress fully, you can stream the data with zcat and extract text using strings. This approach avoids storage limits while still allowing you to analyze the disk and find hidden data efficiently.